Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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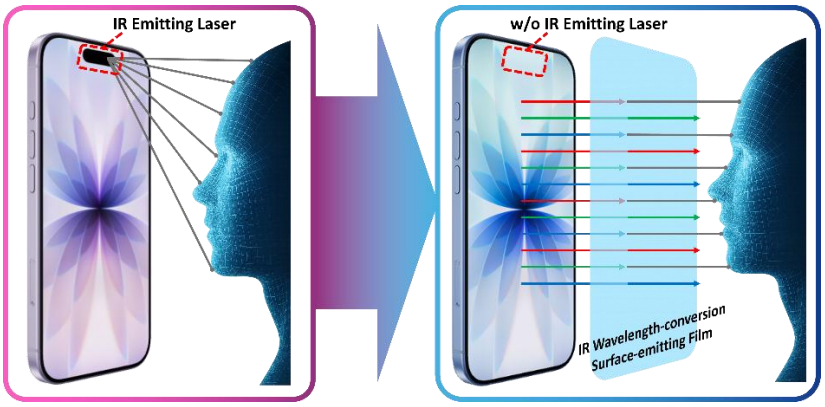
2. Technology Overview

Title of Proposed Technology	External IR(NIR-SWIR) Wavelength-conversion Surface-emitting Film for Interactive Displays (인터랙티브 디스플레이를 위한 외장형 적외선(NIR-SWIR) 파장변환 면발광 필름 개발)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and electronics		Electrical and electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor devices and systems	Sensor element	Display	Display components and Materials
Concept of Proposed Technology	**Core concept**  - Recent display industry has been evolving beyond a simple image-display device into an intelligent interface platform that interacts with users. In various applications such as smartphones, tablets, wearables, and automotive panels,			

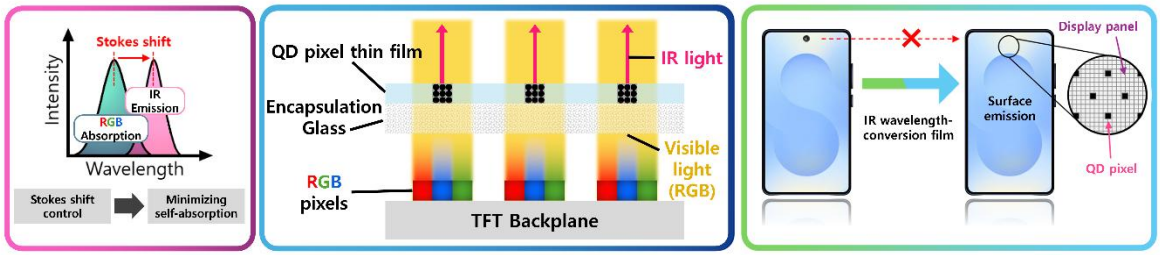
sensor-embedded display (Sensing Display) technology is emerging as a key next-generation innovation area, and for this purpose, the integration of infrared (Infrared) sensing and emission functions is essential.

- Conventional interactive displays utilize externally added light-emitting and light-receiving sensors (VCSELs, photodetectors) separately integrated into the panel. Such a separate module structure causes issues such as space constraints on the front side of the display, design discontinuity, optical misalignment, and increased process complexity. In addition, the efficiency of infrared devices is low.

- This study proposes an infrared surface-emitting film system utilizing display light in order to overcome the above limitations. The film selectively converts visible light (R/G/B) emitted from the display into infrared light (NIR–SWIR), thereby replacing conventional laser-based systems and proposing a new system that enables efficient utilization of infrared light.

- Infrared wavelength-conversion materials are fabricated, and substrate printing technology is applied to the fabricated materials to print fine patterns at the micrometer scale on a transparent substrate without causing issues to display visibility. Each pattern is arranged to be aligned with the corresponding RGB pixels. The fabricated wavelength-conversion film can be placed on the display, and visible light emitted from the display is converted into infrared light and emitted, thereby replacing the role of conventional IR emission systems.

Main functions



[Wavelength-conversion] [Side-view of the display] [Front-view of the display]

- It utilizes a photoluminescence phenomenon in which high-energy visible photons emitted from the display are absorbed and subsequently low-energy infrared photons are emitted. The display is composed of pixels that emit the three primary colors of light—red, green, and blue (RGB)—and the wavelength range of visible light is approximately 380–750 nm.

- Low-bandgap quantum dots (QDs) can absorb the corresponding broadband light (i.e., RGB) and emit charge carriers in the infrared region through radiative recombination. Through this process, a portion of the visible light (380–750 nm) emitted from the display can be converted into infrared light (900–1550 nm). In particular, to achieve high emission efficiency and stability, the QDs are designed in a heterostructure core/shell configuration.

- The infrared photoluminescent pixels performing the above functions are formed on a film substrate. In particular, the pixels are patterned with an ultra-small pixel design to prevent Moiré Effects caused by optical interference with the display and to avoid image quality distortion due to such interference.

- The infrared photons generated in this manner can subsequently be absorbed by an internal optical sensor system of the device and converted into signals, enabling utilization in various fields such as biomedical and security applications depending on the intended purpose.

Potential uses of the proposed technology

- Biometric Authentication & Security
  · Use for biometric identification of users of large- and small-scale equipment
  · Use of the full front surface of the display for facial recognition technology in mobile devices

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- Smart Vehicle (Use in automotive digital clusters)
  · Monitoring of driver status, such as detection of drowsy driving and biometric signal sensing
  · Monitoring of in-vehicle conditions, such as detection of the presence of pets or infants

- Smart home (Biometric signal monitoring)
  · Monitoring of user biometric signals (heart rate, blood flow, oxygen saturation, etc.)
  · Indoor monitoring systems (intrusion detection, resident status monitoring, etc.)

- Entertainment
  · User eye-tracking systems in VR/AR devices
  · Activities utilizing user motion recognition

Rationale for Joint R&D Project

Through this research project, the aim is to develop fabrication technology for an IR QD-based wavelength conversion surface-emitting film.

- The fabrication of the IR wavelength-conversion surface-emitting film proceeds in the following sequence:
  1. Fabrication of wavelength-conversion materials
  2. Ligand exchange of the materials to improve printability
  3. Ink formulation for printing
  4. Pattern printing using the formulated ink
  5. Film fabrication
  6. Evaluation of optical properties and sensing functions

- The research institute (Fraunhofer IAP) primarily conducts research on film fabrication, printing, and coating using nanomaterials and possesses the relevant equipment, technologies, and experience; therefore, it has expertise in processes #3–5 of the target film fabrication process.

- However, the institute currently has limitations in the research on the required materials (processes 1 and 2) and in the evaluation of product properties (process 6) necessary for the proposed technology.

- Therefore, among organizations that are currently producing and conducting R&D on IR wavelength-conversion materials, there is a need for partner institutions (Korean company) that can develop IR wavelength-conversion materials suitable for use in printing processes, supply them to Fraunhofer IAP, and conduct characterization and functional evaluation of the fabricated prototypes.

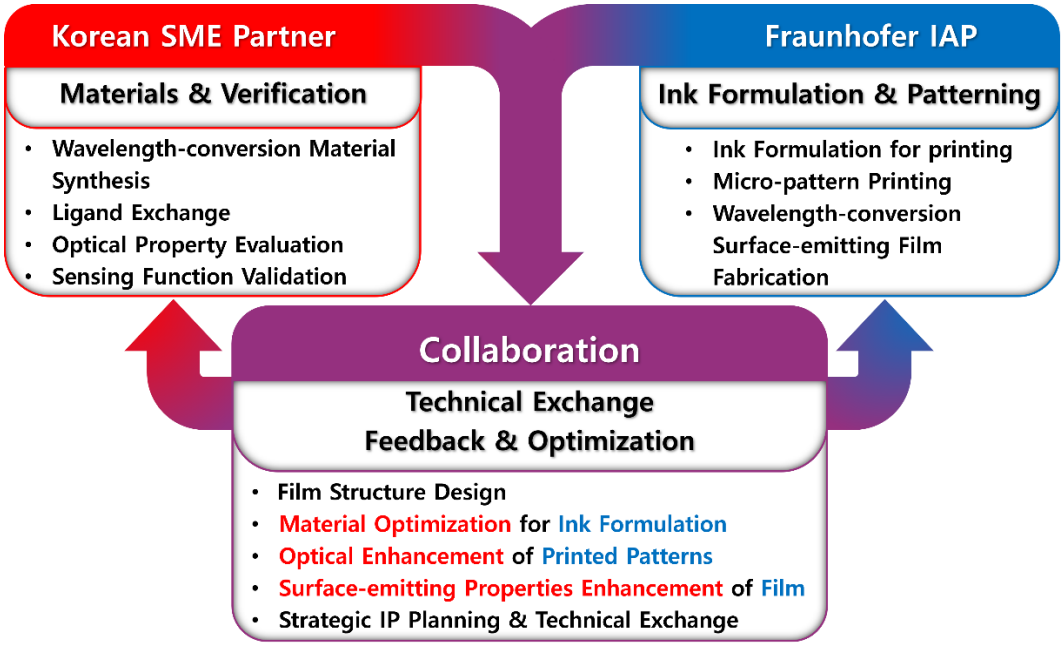
- The fabrication results of the IR wavelength-conversion surface-emitting film are not independent across individual

processes but are interdependent. Above all, the development of the wavelength-conversion materials is the most critical step, and subsequent improvement of material properties for the ink formulation process is essential; in this process, collaboration between material development experts and printing experts, integrating their technologies and experience, is important.

- As a specific example, since the ligand structure of the materials directly affects the viscosity, conductivity, and dispersibility of the ink, the research on ligand exchange of the materials for printing is required. In addition, joint research is needed on the analysis of wavelength-conversion properties of the materials resulting from the ink formulation process.

- As outcomes are derived through international joint research, this also contributes to securing a broad range of intellectual property at the international level. In this process, technologies related to materials are mainly secured by the partner institution (Korean company), while technologies related to printing are mainly secured by the research institute (Fraunhofer IAP), and it also facilitates the acquisition of joint patents and international patents related to ink formulation and related processes.

- Ultimately, through international joint research between Fraunhofer IAP and Korean companies, performance data sharing and international joint demonstration are carried out, and based on this, technological reliability is jointly verified and secured. Furthermore, by deriving technological and business cooperation synergies, it becomes possible to enhance the credibility of Korean material technologies and to establish a foundation for entry into the European and global markets.



Key Words	1. Interactive display 2. IR quantum dot 3. Wavelength-conversion materials 4. Nanomaterial printing 5. Micro-patterning

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	**Development of an external infrared (NIR–SWIR) wavelength-conversion surface-emitting film for displays** 1. Improvement of IR wavelength-conversion QD material technology • Development of Pb-free IR wavelength-conversion QDs synthesis technology • Band structure design for target operating wavelengths • Introduction of a core/shell structure for trap passivation and maximization of photoluminescence • Securing material stability 2. Development of transparent substrate–based IR wavelength-conversion film fabrication technology • Development of QD ink formulation technology for transparent substrates • Development of surface ligand exchange technology • QD thin-film coating and optimization of optical properties • Design to minimize optical interference with RGB displays 3. Implementation of micro-patterned IR wavelength-conversion pixel structure Development of QDs ink formulation technology for transparent substrates • Development of surface ligand exchange technology • QDs thin-film coating and optimization of optical properties • Design to minimize optical interference with RGB displays 4. Design of external display structure and verification of sensing functions • Structural design of an external display wavelength-conversion surface-emitting film • Verification of IR reflected-light–based sensing functions (depth, gesture, biometric signals)			
R&D Target Specifications	• Emission wavelength conversion: 900–1550 nm • PL quantum yield: > 80% • IR wavelength-conversion QD pixel array: ≥ 60 x 60 mm substrate • IR wavelength-conversion QD pixel size: ≤ 10 μm • Fabrication of a wavelength-conversion surface-emitting film prototype • Long-term stability test: after exposure for 500 h at 85 °C and 85% R.H., maintain ≥ 80% intensity when operating under the same display light source • Need to propose practical use-case examples such as distance, bio, and gesture recognition			
R&D Plan	**Preliminary study (Phase 1: 6 months)** Risk analysis of IR wavelength-conversion QD–based core technologies and establishment of the film structure concept/specifications 1. Review of prior technologies and applicability (Risk-oriented) • Investigation of relevant prior studies and confirmation of applicability • Identification of key technical risks of this project's technology compared to existing technologies in terms of performance limitations, process compatibility, and stability 2. Establishment of an IR wavelength-conversion QD material library • Organization of a database of characteristics and result data according to IR wavelength-conversion QD materials and structures • Development of basic synthesis protocols and verification of optical properties			

3. Concept design of a transparent substrate–based wavelength-conversion surface-emitting film
- Selection of transparent substrate materials applicable to displays
- Design of a reference QD ink formulation

4. Securing intellectual property and establishing a patent strategy
- Establishment of an IP portfolio based on materials/film and structure/process

5. Korea–Germany technical exchange
- Strengthening technical understanding and practical discussions through mutual on-site visits
- Operation of working-level meetings and workshops between engineers on both sides
- Establishment of a roadmap for joint research and technical cooperation

Main research (Phase 2: 12 + 12 + 6 = 30 months)
Development of high-efficiency wavelength-conversion surface-emitting films, patterning, large-area fabrication, and sensing demonstration

1. Advancement of IR wavelength-conversion QD material development
- Utilization of the IR wavelength-conversion QD material library secured during the preliminary study period
- Core/shell structure design and maximization of photoluminescence
- Optimization of surface passivation and development of ligand exchange technology
- Establishment of ligand systems suitable for ink fabrication
- Evaluation and improvement of long-term material stability

2. Development of fabrication processes for wavelength-conversion surface-emitting films
- Study of printing process conditions for small-area and large-area fabrication on transparent substrates
- Design to minimize optical interference with RGB displays
- Evaluation of printing results and process suitability using piezo inkjet and EHD-jet equipment (depending on the printing system)
- Study of required physical properties (viscosity, conductivity, concentration, etc.) of QD inks suitable for the selected equipment
- AI-based (Machine Learning–driven) optimization of ink formulation and printing process with automated process window exploration
- Verification and optimization of micro-scale QD ink printing conditions (e.g., substrate–ink interfacial adhesion depending on thermal treatment conditions)
- Verification of IR wavelength-conversion performance and visibility according to QD pixel printing thickness
- Design of QD pixel patterning considering aperture ratio and display color coordinates
- Implementation of uniform QD pixel patterns on large-area substrates
- Study of encapsulation materials and methods for long-term stability

3. Fabrication and performance evaluation of small-area and large-area prototypes
- Fabrication of small-area prototypes for design verification and large-area prototypes for demonstration
- Evaluation of IR wavelength-conversion optical performance (emission wavelength, intensity, quantum yield, transmittance, etc.)
- Evaluation of sensing functions (depth recognition, gesture recognition, etc.)
- Evaluation of long-term stability (high-temperature/high-humidity tests, light irradiation tests, etc.)

4 Securing intellectual property
- Patent application and registration in Korea and Germany

5. Korea–Germany technical exchange
- On-site visits to verify the development process of “materials–film fabrication–validation”
- Discussion of cooperation and improvement measures through working-level meetings

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Research Contents		Phase 1	Phase 2				
		6 Month (M)	6M	6M	6M	6M	6M
Review of prior technologies and applicability, and risk analysis							
IR materials	Verification of optical properties according to material type						
	Material optimization						
	Design and fabrication of core/shell structures						
	Ligand exchange and optimization for ink fabrication						
	Evaluation and improvement of long-term material stability						
Film fabrication	Selection of transparent substrate materials						
	Optimization of physical properties of printing inks						
	Optimization of micro-scale printing conditions						
	AI-based Micro-patterning and optimization of optical properties						
	Large-area wavelength-conversion pixel patterning						
Prototypes	Small-area fabrication and optical performance evaluation						
	Large-area fabrication and sensing function evaluation						

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. Design and fabrication of IR wavelength-conversion QD films • Design of transparent film–based IR wavelength-conversion QD pixel structures • Design to minimize optical interference with RGB displays • Research on IR wavelength-conversion QD ink formulation • Research on, and provision of information on, physical property requirements (viscosity, conductivity, concentration, etc.) of inks suitable for printing equipment • Development of printing processes considering substrate and ink characteristics • Optimization of the film's optical properties (transmittance, color coordinates, etc.) • AI-based (Machine Learning–driven) Optimization of QD Pixel Patterning Process • Study of printing process conditions for small-area and large-area fabrication • Fabrication and provision of prototype films 2. Development of micro-patterning technologies • Development of fine micro-patterning processes for IR wavelength-conversion QD pixels • Study of printing characteristics results according to process conditions • Securing alignment accuracy and uniformity of fine patterns • Fabrication of large-area pixel arrays 3. Development of technologies to secure stability of IR wavelength-conversion films via thin-film coating • Development of thin-film encapsulation processes compatible with IR wavelength-conversion QD pixel layers/transparent substrates • Verification of encapsulation stability and process optimization
Expected R&R of Korean SME Partner	1. Development and advancement of IR wavelength-conversion QD materials • Bandgap tuning of materials to match target conversion wavelengths (900–1550 nm) • Development of Pb-free IR wavelength-conversion QD material synthesis processes • Securing core/shell structure formation and surface passivation technologies • Research on ligand exchange required for ink formulation and printing, and provision of materials • Evaluation and optimization of optical properties and uniformity of the fabricated materials • Evaluation and improvement of long-term material stability 2.Fabrication and performance evaluation of small-area and large-area prototypes • Fabrication of prototypes using the provided IR wavelength-conversion films • Fabrication of small-area prototypes for design verification and large-area prototypes for sensor performance demonstration • Evaluation of IR wavelength-conversion optical performance (emission wavelength, intensity, quantum yield, transmittance, etc.) • Evaluation of sensing functions (depth recognition, gesture recognition, etc.) • Evaluation of long-term stability (high-temperature/high-humidity tests, light irradiation tests, etc.) 3. Securing intellectual property • Domestic and international patent application and registration • Establishment of a material-based IP portfolio • Establishment of a joint patent strategy 4. Market application and commercialization strategy development

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Implementation of interactive displays • The display itself serves as a broadband light source, eliminating the need for external modules • Enables realization of a complete full-screen display without front camera holes or sensor windows 2. Expansion of sensing functions (Beyond Imaging) • NIR–SWIR light exhibits high penetration into skin, blood, and tissue, enabling improved accuracy in biometric signal measurements (SpO_2, heart rate, blood flow velocity, etc.) and non-contact interactions, as well as recognition of a broader range of biological parameters • Enables direct integration of depth, gesture, and biometric recognition functions into existing RGB displays • Light with wavelengths above 1400 nm is safe for vision, posing no issues for long-term exposure 3. Response to future interface markets • Global IT companies are competitively promoting Invisible Sensor Integration technologies for AR/VR, smartphones, and automotive displays. Early acquisition of leading technologies in this field can secure leadership in building a future "display-centric interface ecosystem" • This project provides a foundational technology enabling displays to evolve into interactive platforms capable of sensing and recognition, and will serve as an essential turning point for integrated and immersive display technologies 4. European market and industrial/economic impact • This technology can be expanded into various optical and sensing application markets in Europe and can be widely applied in display, healthcare, automotive electronics, and smart device sectors • In particular, environmentally friendly NIR–SWIR material technologies are evaluated as strategically important core material technologies in the European market, where environmental regulations are stringent • Joint performance verification and international joint demonstration with Fraunhofer IAP, based on materials and process technologies, secure technical reliability and credibility of Korean material technologies • Based on this, it contributes to entry into the European market, establishment of local partnerships, and expansion of mid- to long-term commercialization opportunities

6. References

(1) *Nat. Commun.*, 16, 1666 (2025)

(2) *Adv. Mater. Technol.*, 2401960 (2025)

(3) *Nat. Commun.*, 11, 2874 (2020)

(4) *Small*, 21, 2412420 (2025)

(5) *J. Am. Chem. Soc.,* 142, 3449 (2020)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization